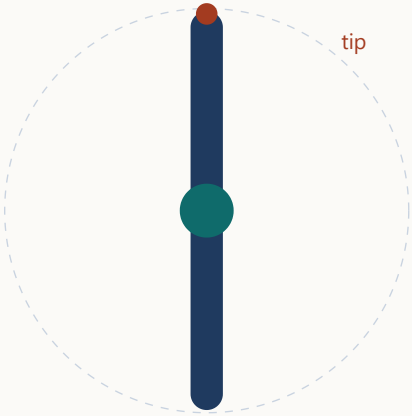
# How fast is the tip of a propeller?

A Cessna 310 propeller is **81 inches** across and turns at **2200 rpm**.

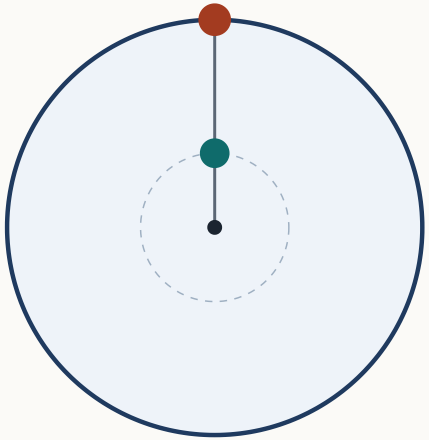
Guess: how fast is the tip moving, in miles per hour?




## About **530 mph**.

By the end of class you can get that number yourself, and do it on Quiz 9 (Wednesday) and Test 2 (Friday).

# Same turn, different speed



- Both dots turn through the **same angle** in the same time: the same **angular velocity** $\omega$.
- The outer dot travels a **bigger circle** in that time, so it moves faster: a bigger **linear velocity** $v$.
- $\omega$ = how fast it **turns** (radians per unit of time).
  $v$ = how fast a point **moves** (distance per unit of time).

# Predict

Ana sits **2 m** from the center of a merry-go-round. Ben sits **6 m** from the center. It turns once every **10 s**.

Question 1

Who has the larger angular velocity?

**Neither. Both turn 2π rad in 10 s: ω = 2π/10 ≈ 0.628 rad/s.**

Question 2

Who has the larger linear velocity, and how many times larger?

**Ben, 3 times larger: 6 m is 3 × 2 m. (Ana ≈ 1.26 m/s, Ben ≈ 3.77 m/s.)**

# Where $v = r\omega$ comes from

$$v = {}^{s}/_{t}$$

On a circle, the distance traveled is an arc: $s = r\theta$ ($\theta$ in radians).

$$v = {}^{r\theta}/_{t} = r \cdot {}^{\theta}/_{t} = r\omega$$

Linear velocity = radius × angular velocity, with $\omega$ in **radians** per unit of time. Radians have no units, so (ft)(rad/min) = ft/min.

# The unit chain: five lines, every time

1. **Radius.** Given a diameter? Divide by 2.
2. **Angular velocity in radians.** rpm × 2π rad per revolution.
3. **Multiply.** $v = r\omega$
4. **Convert units.** 12 in = 1 ft · 5280 ft = 1 mi · 60 min = 1 h · 1000 m = 1 km · 3600 s = 1 h
5. **Round once, at the end,** and write the units.

Shortcut for rpm: $v = (\text{rpm})(2\pi r)$ gives distance per minute. It is the same as lines 2 and 3 together.

WORKED EXAMPLE

# Riverboat paddle wheel

The paddles have a radius of **8.50 ft** and turn at **20.0 rpm**. How fast is a paddle tip moving, in ft/min and mi/h?

| | |
|---|---|
| 1 · Radius | $r = 8.50\ \text{ft}$ |
| 2 · Radians per minute | $\omega = 20.0\ {}^{\text{rev}}/_{\text{min}} \cdot {}^{2\pi\ \text{rad}}/_{\text{rev}} = 40\pi\ \text{rad/min}$ |
| 3 · Multiply | $v = r\omega = 8.50(40\pi) = 340\pi \approx 1068.1\ \text{ft/min}$ |
| 4–5 · Convert, round | $1068.1\ {}^{\text{ft}}/_{\text{min}} \cdot {}^{60\ \text{min}}/_{1\ \text{h}} \cdot {}^{1\ \text{mi}}/_{5280\ \text{ft}} \approx 12.1\ \text{mi/h}$ |

# Hang glider

A hang glider flies a circular arc of radius **90.0 m** with angular velocity **0.125 rad/s**. Find its speed in m/s and km/h.

| | |
|---|---|
| 1 · Radius | $r = 90.0\ \text{m}$ |
| 2 · Radians per second | $\omega = 0.125\ \text{rad/s}$ (already in radians) |
| 3 · Multiply | $v = r\omega = 90.0(0.125) = 11.25\ \text{m/s}$ |
| 4–5 · Convert, round | $11.25\ {}^{\text{m}}/_{\text{s}} \cdot {}^{3600\ \text{s}}/_{1\ \text{h}} \cdot {}^{1\ \text{km}}/_{1000\ \text{m}} = 40.5\ \text{km/h}$ Round 11.25 to 11.3 first and you get 40.7, which is wrong. |

YOUR TURN, WITH A HEAD START · WORK WITH A PARTNER

# Back to the propeller

Diameter **81 in**, turning at **2200 rpm**. Find the tip speed in ft/min and mi/h.

| | |
|---|---|
| 1 · Radius (done for you) | $r = 81 \div 2 = 40.5$ in |
| 2 · Radians per minute | $\omega =$ _____ $= 2200 \cdot 2\pi = 4400\pi$ rad/min |
| 3 · Multiply | $v =$ _____ in/min $= 40.5(4400\pi) \approx 559{,}832$ in/min |
| 4–5 · Convert, round | _____ ft/min and _____ mi/h $\approx 46{,}652.7$ ft/min $\approx 530.1$ mi/h |

SPOT THE ERROR

# Three wrong answers, one right one

A wheel **2.0 ft** across turns at **300 rpm**. Find the rim speed in ft/min.

Answer A

$v = 2.0(300)(2\pi) \approx 3769.9 \text{ ft/min}$

**Used the diameter. The answer is doubled.**

Answer B

$v = 1.0(300)(360) = 108{,}000 \text{ ft/min}$

**Used degrees. ω must be in radians.**

Answer C

$v = 1.0(300) = 300 \text{ ft/min}$

**Left ω in rpm (forgot 2π per revolution).**

Answer D

$v = 1.0(300)(2\pi) = 600\pi \approx 1885.0 \text{ ft/min}$

**Correct: radius, radians, multiply.**

YOU TRY · ON YOUR OWN

# Watch second hand (worksheet 12a)

The sweep second hand of a watch is **21.0 mm** long. It makes one turn per minute.

Find the linear velocity of its tip in **mm/s** and in **cm/min**. Use the five lines.

1. Radius
2. Radians per second
3. Multiply
4. Convert
5. Round and label

CHECK YOUR WORK, LINE BY LINE

# Watch second hand: answer

| | |
|---|---|
| 1 · Radius | $r = 21.0\ \text{mm}$ |
| 2 · Radians per second | $\omega = {}^{2\pi\ \text{rad}}/_{60\ \text{s}} = {}^{\pi}/_{30}\ \text{rad/s}$ |
| 3 · Multiply | $v = 21.0 \cdot {}^{\pi}/_{30} = 0.7\pi \approx 2.2\ \text{mm/s}$ |
| 4–5 · Convert, round | $0.7\pi\ {}^{\text{mm}}/_{\text{s}} \cdot {}^{60\ \text{s}}/_{1\ \text{min}} \cdot {}^{1\ \text{cm}}/_{10\ \text{mm}} \approx 13.2\ \text{cm/min}$ |

STRETCH · TEST 2 STYLE

# Wind turbine

The blades of a GE 1.5s wind turbine are **116 ft** long and turn at **22.0 rpm**.

a) Tip speed in ft/min and mi/h

$r = 116$ ft (the blade is the radius)

$v = 116(22.0)(2\pi) = 5104\pi \approx 16{,}035$ ft/min

$\approx 182.2$ mi/h

b) Area swept by the blades, in acres (1 acre = 43,560 ft²)

$A = \pi r^2 = \pi(116)^2 \approx 42{,}273$ ft²

$42{,}273 \div 43{,}560 \approx 0.970$ acre

# This week

**Wednesday**

## **Quiz 9: circles**, in class on paper

- degrees to radians
- circumference, arc length, sector area
- central and inscribed angles
- a volume word problem

Practice: Quiz 9 practice (PDF) on the course page

**Friday**

## **Test 2**, in class on paper

- Printed: compound and continuous interest, Heron's formula
- **From memory:** $v = r\omega$, rpm × $2\pi$, arc length, sector area, circle formulas

Practice: Test 2 practice (PDF) and the Test 2 review page

EXIT TICKET · HAND IN BEFORE YOU LEAVE

# Box fan

A box fan's blades are **24 in** across and turn at **1500 rpm**.

a) What is the radius, in feet?

b) What is the linear speed of a blade tip, in ft/min?

c) What is that speed in mi/h?

Show all five lines of the unit chain. Round only at the end.